7.12 Constrained Langevin Dynamics

We may use a splitting approach to develop discretization algorithms for other forms of dynamics. In Chap. 4, we included holonomic constraints to the ODEs defining constant-energy Hamiltonian dynamics, as a way to increase the stability of our algorithms and increase the usable timestep. We can add constraints to the Langevin dynamics SDEs similarly as

$$\mathrm{d}\boldsymbol{q} = \boldsymbol{M}^{-1}\boldsymbol{p}\mathrm{d}t, \tag{7.43}$$

$$\mathrm{d}\boldsymbol{p} = -\nabla U(\boldsymbol{q})\mathrm{d}t - \gamma\boldsymbol{p}\mathrm{d}t + \sqrt{2\gamma k_{\mathrm{B}}\mathrm{T}\boldsymbol{M}}\mathrm{d}\mathbf{W} - \boldsymbol{g}'(\boldsymbol{q})^T\boldsymbol{\lambda}\,\mathrm{d}t, \tag{7.44}$$

$$\mathbf{0} = \boldsymbol{g}(\boldsymbol{q}), \tag{7.45}$$

where the vector of l constraints is given by $\boldsymbol{g} : \mathbb{R}^{\mathrm{N_c}} \to \mathbb{R}^l$ and $\boldsymbol{\lambda}$ is an l-dimensional vector of Lagrange multipliers. $\mathbf{W}$ is a vector of standard Wiener processes, with the usual friction parameter $\gamma > 0$. As in the constant energy dynamics, permissible phase points lie in the co-tangent bundle defined by the constraint and hidden constraint

$$\boldsymbol{g}(\boldsymbol{q}) = \mathbf{0}, \qquad \boldsymbol{g}'(\boldsymbol{q})\boldsymbol{M}^{-1}\boldsymbol{p} = \mathbf{0},$$

while in the infinite time limit, we have that solution trajectories of constrained Langevin (7.43)–(7.45) sample $\tilde{\rho}_\beta$ ergodically, such that

$$\lim_{T\to\infty}\frac{1}{T}\int_0^T \phi(\boldsymbol{q}(t),\boldsymbol{p}(t))\mathrm{d}t = \int_{\mathcal{D}} \phi(\boldsymbol{q},\boldsymbol{p})\tilde{\rho}_\beta \mathrm{d}\omega,$$

where $\tilde{\rho}_\beta$ is the normalized constrained canonical distribution

$$\tilde{\rho}_\beta = Z^{-1}\exp(-\beta H(\boldsymbol{q},\boldsymbol{p}))\delta[\boldsymbol{g}(\boldsymbol{q})]\delta[\boldsymbol{g}'(\boldsymbol{q})\boldsymbol{M}^{-1}\boldsymbol{p}],$$

for normalizing constant Z and where $\delta[\boldsymbol{z}] := \prod_i \delta[z_i]$ for a vector $\boldsymbol{z}$. The goal is now to find solutions to the constrained dynamics, in order to sample the target distribution.

One approach for algorithms is simply to evolve unconstrained Langevin dynamics using any algorithm of choice, then follow each step by a step of the SHAKE algorithm (see Chapter 4 or [322]) to pull the system back on to the manifold satisfying the holonomic constraints.